The area of a rhombus and of a trapezium

Two formulas, both of them the parallelogram formula in disguise.

LESSON 49 | 1 × 40 MIN | GEOMETRY 8, ТЕМЫ 49–50 | STAGE 9 · 7.2

LESSON CLOCK

5' | 12' | 14' | 5' | 4'

Recall the rhombus properties	5 min
Area of a rhombus	12 min
Area of a trapezium	14 min
Working backwards	5 min
Homework	4 min

LEARNING OBJECTIVES

- Use $S = ½\, d_1 d_2$ for a rhombus.
- Use $S = ½\, (a + b)\, h$ for a trapezium.
- Explain why each formula works.
- Find a missing length from a given area.

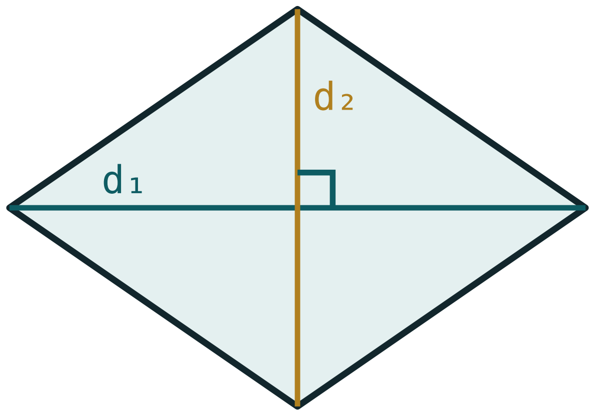
TEXTBOOK

National	Темы 49–50, pp. 114–118
Cambridge	Learner's Book pp. 149–155
Workbook	Workbook 7.2

01 Explanation

The rhombus

A rhombus is a parallelogram, so $S = a \cdot h$ still works. But its diagonals are perpendicular and bisect each other, and that gives a second formula which is usually more convenient:



$$S = \tfrac{1}{2} \cdot d_1 \cdot d_2$$

The perpendicular diagonals cut the rhombus into four congruent right triangles.

AREA OF A RHOMBUS

$$S = \frac{1}{2}\, d_1 \cdot d_2$$

Why: the four right triangles each have legs $\frac{d_1}{2}$ and $\frac{d_2}{2}$, so each has area $\frac{d_1 d_2}{8}$, and four of them make $\frac{d_1 d_2}{2}$.

The same formula holds for any quadrilateral whose diagonals are perpendicular — a kite, for instance — and for a square, where $d_1 = d_2$.

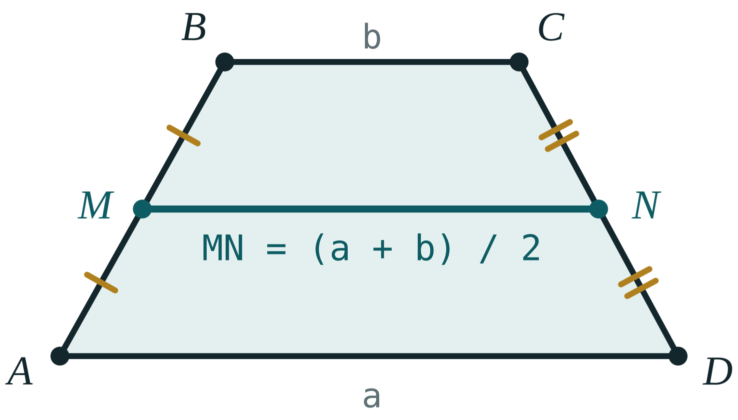
The trapezium

Copy the trapezium, turn the copy through $180°$ and join the two along a leg. The result is a parallelogram of base $a + b$ and height h. The trapezium is half of it:

AREA OF A TRAPEZIUM

$$S = \frac{a + b}{2} \cdot h$$

— the average of the two parallel sides, times the height.



The midline of a trapezium is the average of the two parallel sides, so the area is midline × height.

Since the midline $m = \frac{a+b}{2}$, the formula reads simply $S = m \cdot h$: area is midline times height, exactly as for a rectangle.

AVERAGE, THEN MULTIPLY

The commonest error is $(a + b) \cdot h$ without the halving — an answer exactly twice too big. Sanity-check against the enclosing rectangle: a trapezium must be smaller than $b \cdot h$ when b is the longer parallel side.

02 Worked examples

EXAMPLE 1 **A rhombus has diagonals *12 cm* and *16 cm*. Find its area, its side and its height.**

1. $S = \frac{1}{2} \cdot 12 \cdot 16 = 96\ cm^2$
2. *half-diagonals 6 and 8*
3. $a = \sqrt{36 + 64} = 10\ cm$

   Pythagoras in one of the four right triangles.
4. $96 = 10 \cdot h,\ h = 9.6\ cm$

   Using the parallelogram formula.

ANSWER

$96\ cm^2$, side $10\ cm$, height $9.6\ cm$

EXAMPLE 2 **A trapezium has parallel sides *8 cm* and *14 cm* and height *5 cm*. Find its area and its midline.**

1. $m = \frac{8 + 14}{2} = 11\ cm$

   The midline.
2. $S = 11 \cdot 5$
3. $= 55\ cm^2$

   Less than $14 \cdot 5 = 70$ ✓

ANSWER

$55\ cm^2$, midline $11\ cm$

EXAMPLE 3 **A trapezium has area $84\ cm^2$, height $7\ cm$ and one parallel side $9\ cm$. Find the other.**

1. $$84 = \frac{9 + b}{2} \cdot 7$$

   Substitute what you know.

2. $$12 = \frac{9 + b}{2}$$

   Divide by 7.

3. $$24 = 9 + b$$

   Multiply by 2.

4. $$b = 15\ cm$$

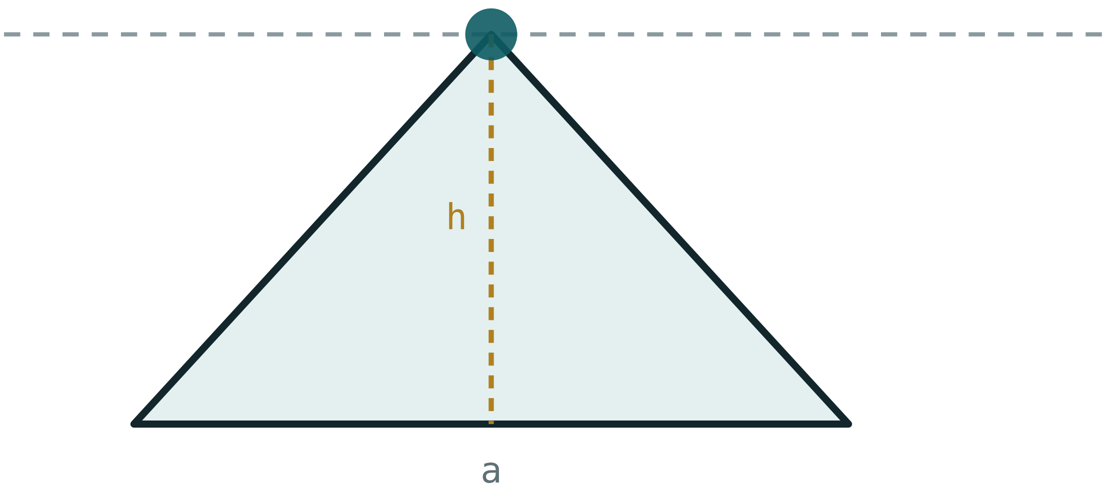
ANSWER

$15\ cm$

03 Interactive model

Change the base and height and check each formula against the readout.

Base, height and area



base a **220** | height h **120** | area ½ah **13200** | perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Rhombus	Romb	Ромб
Diagonal	Diagonal	Диагональ
Trapezium	Trapetsiya	Трапеция
Parallel sides	Parallel tomonlar	Основания трапеции
Midline	O‘rta chiziq	Средняя линия
Height of a trapezium	Trapetsiya balandligi	Высота трапеции
Kite	Deltoid	Дельтоид
Perpendicular diagonals	Perpendikulyar diagonallar	Перпендикулярные диагонали

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A rhombus with diagonals 10 and 8 has area:

- 80
- 40
- 18
- 20

A trapezium with parallel sides 6 and 10 and height 4 has area:

- 64
- 32
- 40
- 16

The midline of a trapezium with parallel sides 7 and 13 is:

- 20
- 10
- 6
- 91

A square of diagonal 6 has area:

- 36
- 18
- 12
- 9

A rhombus of area 60 has one diagonal 10. The other is:

- 6
- 12
- 120
- 30

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. *A rhombus has diagonals 6 and 8. Find its area.*

   ANSWER *24*

2. *A trapezium has parallel sides 5 and 9 and height 4. Find its area.*

   ANSWER *28*

3. *Find the midline of a trapezium with parallel sides 4 and 10.*

   ANSWER *7*

4. *A square has diagonal 10. Find its area.*

   ANSWER *50*

5. *A rhombus has diagonals 14 and 10. Find its area.*

   ANSWER *70*

6. *A trapezium has midline 8 and height 5. Find its area.*

   ANSWER *40*

7. *A rhombus has area 48 and one diagonal 12. Find the other.*

   ANSWER *8*

Medium · the standard question

7 PROBLEMS

1. *A rhombus has diagonals 16 and 30. Find its area and its side.*

   ANSWER *240*, side *17*

2. *A trapezium has area 96, height 8 and one parallel side 10. Find the other.*

   ANSWER *14*

3. *A rhombus of side 13 has one diagonal 24. Find its area.*

   ANSWER other diagonal *10*, area *120*

4. *A trapezium has parallel sides 12 and 18 and area 105. Find its height.*

   ANSWER *7*

5. *A kite has perpendicular diagonals 9 and 14. Find its area.*

   ANSWER *63*

6. *A rhombus has side 10 and height 8. Find its area.*

   ANSWER *80*

7. *A trapezium has midline 11 and area 77. Find its height.*

   ANSWER *7*

Hard · stretch and reason

7 PROBLEMS

1. *A rhombus has perimeter 40 and one diagonal 12. Find its area.*

   ANSWER side 10, other diagonal 16, area 96

2. *An isosceles trapezium has parallel sides 10 and 22 and legs 10. Find its area.*

   ANSWER $h = 8$, area 128

3. *A trapezium has parallel sides in the ratio 2 : 5, height 6 and area 63. Find both sides.*

   ANSWER 6 and 15

4. *A rhombus of area 120 has side 13. Find both diagonals.*

   ANSWER 10 and 24

5. *Show that the formula for a trapezium gives the parallelogram formula when a = b.*

   ANSWER $S = \frac{a + a}{2} \cdot h = a \cdot h$ — the trapezium becomes a parallelogram.

6. *A trapezium is divided by a diagonal into two triangles. Find the ratio of their areas if the parallel sides are 6 and 10.*

   ANSWER $3 : 5$ — the two triangles share the height

7. *The diagonals of a quadrilateral are perpendicular, of lengths 12 and 9. Must its area be 54?*

   ANSWER Yes — cutting along the diagonals gives four right triangles whose areas sum to $\frac{1}{2} d_1 d_2$, whatever the quadrilateral's shape, provided the diagonals meet inside it.

07 Homework

Homework — 5 problems

Geometry 8, Темы 49–50, pp. 114–118. Check every trapezium answer against $b \cdot h$.

1. *A rhombus has diagonals 18 and 24. Find its area and its side.*
2. *A trapezium has parallel sides 7 and 15 and height 6. Find its area.*
3. *A trapezium has area 120, height 10 and one parallel side 9. Find the other.*

4. *A square has diagonal 14. Find its area.*

5. *A rhombus has perimeter 52 and one diagonal 24. Find its area.*